FRC Sensor CAN · Revision B purchasing BOM

51 order lines · 144 fitted components + 10 shunts · Indicative total **US$59.30**

Dated observations: 2026-09-07/08. Quantity-one prices are applied to every unit; actual quantity breaks may reduce cost. PCB, stencil, tools, mounting hardware, shipping and tax are excluded. This is not a live quote or reserved inventory. Recheck exact order codes before purchase.

Use DigiKey_import.csv to import an order. This PDF lists exact assembly quantities. Full ratings, footprint IDs, sources and alternative-review notes are in BOM_grouped.csv. The interactive BOM provides placement guidance.

References	Qty	Value / rating	Manufacturer part number	DigiKey order code	Observed stock	Unit USD	Line USD
F1	1	1.5A FAST	**045101.5MRL** Littelfuse	F2579CT-ND	7,870	2.47	2.47
J1, J2, J3	3	2P PUSH 3.5mm	**1824420000** Weidmuller	281-2020-ND	12,805	1.54	4.62
J4, J5, J6	3	3P PUSH 3.5mm	**1824430000** Weidmuller	281-2021-ND	1,698	2.21	6.63
Y1	1	8MHz 18pF	**ABM3B-8.000MHZ-B2-T** Abracon	535-9720-1-ND	70,541	0.76	0.76
Q1	1	AO4485	**AO4485** Alpha & Omega Semiconductor	785-1551-1-ND	64,905	1.51	1.51
U2	1	AP2112K-3.3	**AP2112K-3.3TRG1** Diodes Incorporated	AP2112K-3.3TRG1DICT-ND	159,040	0.28	0.28
D102, D103, D202, D203, D302, D303	6	BAT46W	**BAT46W-E3-18** Vishay	BAT46W-E3-18GICT-ND	19,931	0.52	3.12
C25, C26	2	30pF / 50V / 5%	**C0805C300J5GACTU** KEMET	399-C0805C300J5GACTUCT-ND	9,211	0.33	0.66
C101, C201, C301	3	1nF / 50V / 5%	**CC0805JRNPO9BN102** YAGEO	311-1122-1-ND	105,431	0.15	0.45
C7, C10, C23, C42	4	1uF / 50V / 10%	**CC0805KKX7R9BB105** YAGEO	311-1886-1-ND	112,870	0.32	1.28
C28, C29	2	10nF / 50V / 10%	**CC0805KRX7R9BB103** YAGEO	311-1136-1-ND	360,198	0.11	0.22
C3, C6, C9, C11, C20, C21, C22, C27, C40, C41, C50, C17	12	100nF / 50V / 10%	**CC0805KRX7R9BB104** YAGEO	311-1140-1-ND	1,921,549	0.11	1.32

References	Qty	Value / rating	Manufacturer part number	DigiKey order code	Observed stock	Unit USD	Line USD
C14	1	33nF / 50V / 10%	**CC0805KRX7R9BB333** YAGEO	311-1374-1-ND	24,641	0.14	0.14
C4, C5, C13, C12, C15, C16	6	22uF / 25V / 10%	**CL31A226KAHNNNE** Samsung Electro-Mechanics	1276-3047-1-ND	426,249	0.48	2.88
C2	1	10uF / 50V / 10%	**CL32B106KBJNNWE** Samsung Electro-Mechanics	1276-3388-1-ND	149,003	0.62	0.62
C1	1	47uF / 50V / 20%	**EEE-FK1H470P** Panasonic	PCE3815CT-ND	10,108	0.93	0.93
R2, R107, R207, R307	4	17.8k / 1% / 0.125 W	**ERJ-6ENF1782V** Panasonic Industry	P17.8KCCT-ND	33,381	0.12	0.48
D20	1	ESD2CAN24-Q1	**ESD2CAN24DBZRQ1** Texas Instruments	296-ESD2CAN24DBZRQ1CT-ND	255,809	0.87	0.87
U7	1	LM339LV	**LM339LVDR** Texas Instruments	296-LM339LVDRCT-ND	1,085	0.42	0.42
U1	1	LMR51420XDDCR	**LMR51420XDDCR** Texas Instruments	296-LMR51420XDDCRCT-ND	14,239	1.32	1.32
D10, D12, D13, D14	4	GREEN	**LTST-C170KGKT** Lite-On	160-1414-1-ND	498,450	0.19	0.76
D11	1	RED	**LTST-C170KRKT** Lite-On	160-1415-1-ND	489,805	0.18	0.18
10 configuration shunts	10	2 mm shunt	**M22-1900005** Harwin	952-1305-ND	31,908	0.35	3.50
JP2, JP102, JP202, JP302	4	SELECT	**M22-2010205** Harwin	952-2273-ND	7,644	0.16	0.64
JP101, JP201, JP301	3	SELECT	**M22-2010305** Harwin	952-2274-ND	35,632	0.22	0.66
JP103, JP203, JP303	3	SELECT	**M22-2020305** Harwin	952-2133-ND	838	0.41	1.23
D3	1	12V ZENER	**MMSZ12T1G** onsemi	MMSZ12T1GOSCT-ND	34,429	0.17	0.17
R1, R7, R12, R23, R106, R206, R306	7	100k / 1% / 0.125 W	**RC0805FR-07100KL** YAGEO	311-100KCRCT-ND	1,152,791	0.11	0.77

References	Qty	Value / rating	Manufacturer part number	DigiKey order code	Observed stock	Unit USD	Line USD
R3, R6, R9, R11, R14, R15, R21, R22, R24, R25, R26, R40, R41, R105, R205, R305	16	10k / 1% / 0.125 W	**RC0805FR-0710KL** YAGEO	311-10.0KCRCT-ND	2,321,007	0.10	1.60
R101, R102, R201, R202, R301, R302	6	110k / 1% / 0.125 W	**RC0805FR-07110KL** YAGEO	311-110KCRCT-ND	5,943	0.10	0.60
R4, R13, R30, R31, R32, R33, R34	7	1k / 1% / 0.125 W	**RC0805FR-071KL** YAGEO	311-1.00KCRCT-ND	894,200	0.10	0.70
R103, R203, R303	3	20k / 1% / 0.125 W	**RC0805FR-0720KL** YAGEO	311-20.0KCRCT-ND	27,443	0.10	0.30
R5	1	3.9k / 1% / 0.125 W	**RC0805FR-073K9L** YAGEO	311-3.90KCRCT-ND	131,203	0.10	0.10
R8, R110, R111, R210, R211, R310, R311	7	4.7k / 1% / 0.125 W	**RC0805FR-074K7L** YAGEO	311-4.70KCRCT-ND	564,468	0.10	0.70
R10	1	59k / 1% / 0.125 W	**RC0805FR-0759KL** YAGEO	311-59.0KCRCT-ND	29,786	0.10	0.10
R108, R208, R308	3	7.15k / 1% / 0.125 W	**RC0805FR-077K15L** YAGEO	311-7.15KCRCT-ND	48,097	0.10	0.30
R104, R204, R304	3	910k / 1% / 0.125 W	**RC0805FR-07910KL** YAGEO	YAG3336CT-ND	58,915	0.10	0.30
R20	1	0 / jumper / 0.125 W	**RC0805JR-070RL** YAGEO	311-0.0ARCT-ND	1,733,368	0.10	0.10
R42	1	120 / 1% / 0.25 W	**RC1206FR-07120RL** YAGEO	311-120FRCT-ND	126,570	0.10	0.10
R109, R209, R309	3	4.7k / 1% / 0.75 W	**RC2010FK-074K7L** YAGEO	13-RC2010FK-074K7LCT-ND	7,566	0.23	0.69
D101, D201, D301	3	SMAJ30A	**SMAJ30A** Littelfuse	SMAJ30ALFCT-ND	104,069	0.50	1.50
D2	1	SMBJ22A	**SMBJ22A-E3/52** Vishay	SMBJ22A-E3/52GICT-ND	705	0.56	0.56
L2	1	4.7uH	**SRP4020TA-4R7M** Bourns	SRP4020TA-4R7MCT-ND	9,132	1.07	1.07
L1	1	10uH	**SRP7028A-100M** Bourns	SRP7028A-100MCT-ND	9,366	1.10	1.10

References	Qty	Value / rating	Manufacturer part number	DigiKey order code	Observed stock	Unit USD	Line USD
D4	1	SS14	**SS14-E3/61T** Vishay	SS14-E3/61TGICT-ND	99,297	0.47	0.47
U5	1	STM32F042K6T6	**STM32F042K6T6** STMicroelectronics	497-14647-ND	3,072	4.34	4.34
U6	1	TCAN3413DR	**TCAN3413DR** Texas Instruments	296-TCAN3413DRCT-ND	5,237	1.87	1.87
C8, C24	2	4.7uF / 25V / 10%	**TMK212AB7475KG-T** Taiyo Yuden	587-2990-1-ND	718,393	0.26	0.52
U3	1	TPS1H200A-Q1	**TPS1H200AQDGNRQ1** Texas Instruments	296-50416-1-ND	25,656	1.78	1.78
U4	1	TPS2553DBVR-1	**TPS2553DBVR-1** Texas Instruments	296-39380-1-ND	5,463	1.06	1.06
U8	1	TPS613222A	**TPS613222ADBVR** Texas Instruments	296-50503-1-ND	15,432	0.55	0.55